

SHLOK KUMAR GOENKA

Technical Product Analyst | Product Analytics • AI & Automation • Financial Technology

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PROFESSIONAL SUMMARY

Data Science & Engineering undergraduate (CGPA 8.84/10) and CFA Level I candidate who converts data into product decisions. Build end-to-end analytical solutions: SQL data models, Python pipelines, BI dashboards, predictive models, and RAG / agentic AI automation that eliminates repetitive manual work. Translate ambiguous business requirements into measurable outcomes for technical and non-technical stakeholders alike. Oracle Certified Generative AI Professional; three patents filed with the Indian Patent Office; two national hackathon wins. Seeking a Technical Product Analyst role at the intersection of product, data, AI, and capital markets.

CORE SKILLS

Data Analysis & Statistics: SQL (joins, aggregations, window functions, query optimization), Python (Pandas, NumPy, Scikit-learn), statistical modeling, hypothesis testing, root cause analysis, KPI definition and tracking, trend & anomaly analysis

Reporting, BI & Visualization: Power BI, Tableau, Advanced Excel, Matplotlib, Seaborn, Recharts, JFreeChart, executive dashboards and reporting

Databases & Data Engineering: PostgreSQL, MySQL, MongoDB, SQLAlchemy, Supabase (pgvector), relational data modeling, ETL/data pipelines, data quality validation

AI, ML & Generative AI: LLMs, RAG systems, LangChain, Hugging Face Transformers, PyTorch, TensorFlow, spaCy, NLP, Computer Vision (YOLOv8, DeepSORT, ResNet), predictive modeling, model evaluation & monitoring, Oracle Generative AI

AI Automation & Orchestration: n8n, Make (Integromat), AI agent development, agentic systems, prompt engineering, workflow orchestration, process automation, Google Gemini API, Ollama

Software Engineering: Python, Java, C, R, FastAPI, Flask, Next.js (React), Streamlit, Docker, Git/GitHub, Linux/Bash, OOP, REST APIs, Jupyter

Product & Business: Requirements analysis, product & business metrics, data-driven decision making, stakeholder communication, technical documentation, Agile team delivery, financial markets domain (CFA Level I: equities, fixed income, derivatives, portfolio management, quantitative methods)

ANALYTICS, AI & PRODUCT PROJECTS

C.I.T.A.D.E.L.: Civic Intelligence & Transparency Platform

Hackathon Winner | 2026

- **Architected** a dual-portal (Citizen and Government) product unifying RAG-powered policy navigation, real-time violation detection, and administrative automation into a single exception-based workflow.
- **Engineered** a hybrid RAG assistant (Gemini Pro 1.5 with Llama 3.2 fallback) indexed over 500+ civic procedures, sustaining **80%+ retrieval confidence** and reducing manual document lookup to a single natural-language query.
- **Deployed** a YOLOv8 + DeepSORT computer-vision module reaching **94.7% detection accuracy**, and an EasyOCR document-intelligence pipeline extracting identity fields in **under 1.5 seconds**.
- **Instrumented** anomaly monitoring across civic datasets to automatically surface irregular records for review, plus a deep-learning misinformation classifier, eliminating manual spot-checking.

Stack: React (Vite), FastAPI, Supabase (pgvector), YOLOv8, DeepSORT, EasyOCR, Gemini Pro 1.5, Ollama/Llama 3.2 | github.com/shlokDS16/CITADEL

ARGUS: AI-Powered Digital Forensics Platform

Hackathon Winner | 2026

- **Led** development of a multi-modal analytics pipeline ingesting social media posts, images, and PDFs to automate evidence triage for cyber-harassment and identity-theft investigations.
- **Delivered 92.4% authorship-attribution accuracy** (Burrows' Delta) and **94.8% deepfake-detection accuracy** via a ResNet-GAN ensemble, validated against Midjourney and DALL-E generated samples.
- **Modeled** suspect networks as graphs and applied betweenness centrality (NetworkX) to isolate broker nodes at **85% confidence**, cutting end-to-end case processing to **under 1.5 seconds**.
- **Enforced** a tamper-evident chain of custody using SHA-256 hashing, satisfying evidentiary-integrity and auditability requirements.

Stack: Next.js, FastAPI, spaCy, ResNet-18, NetworkX, Gemini 2.0 API, SHA-256 | github.com/shlokDS16/ARGUS

PrivacyGuard Analytics: Privacy-Preserving Compliance Platform

Academic Project

- **Built** a full-stack analytics platform enforcing k-anonymity and l-diversity on regulated datasets; a JIT + heatmap strategy **reduced privacy violations from 45 to 5 (89% reduction)** while retaining **64.2% data utility** across 60 benchmark queries (UCI Heart Disease dataset).
- **Implemented** an Ed25519-signed, tamper-proof audit trail for HIPAA-aligned regulatory reporting, sustaining **sub-60 ms P95 latency** under load.
- **Containerized** the service with Docker over a PostgreSQL/SQLAlchemy data layer, demonstrating compliance-by-design in a production-shaped architecture.

Stack: Next.js 14, FastAPI, PostgreSQL, SQLAlchemy, Docker, Ed25519 | github.com/shlokDS16/Privacy-guard

Carbon-Aware Cloud Scheduler: Cost & Sustainability Optimizer

Research / Patent-Pending

- **Designed** the patent-pending ACEI multi-objective algorithm to route workloads across 10 global datacenters using real-time carbon-intensity data from 200+ countries (OWID, **14M+ records**).
- **Quantified** impact through CloudSimPlus benchmarking: **15–25% carbon reduction, 8–12% cost improvement, and 5–10% latency optimization**, with automatic failover and SLA compliance.
- **Analyzed** the cost–carbon–latency trade-off surface and visualized results (JFreeChart) to support scheduling decisions; this analysis anchored the patent filing.

Stack: Java 17, CloudSimPlus, Apache Commons CSV, JFreeChart, OWID CO₂ Dataset | Patent App. No. 202641026992

PATENTS & INTELLECTUAL PROPERTY

Three patents filed with the Indian Patent Office (IPR Cell), a distinction rarely held by a pre-final-year undergraduate.

- **Hybrid Multi-Objective Cloud Task Scheduling System for Sustainable Computing Infrastructure:** carbon-aware workload distribution across global datacenters with automatic failover and SLA compliance. (App. No. 202641026992)
- **Hardware-Interfaced Spatio-Temporal Control Gating Apparatus for Agricultural Machinery:** cyber-physical control system integrating temporal gating with adaptive actuator enforcement for precision agriculture. (App. No. 202641054903)
- **Quantum-Assisted Longitudinal State Transition Engine for Counterfactual Progression Modeling:** quantum-inspired engine for longitudinal state-transition analysis and counterfactual scenario generation in dynamic systems. (App. No. 202641054904)

CERTIFICATIONS

- **Oracle Certified Generative AI Professional**, Oracle (2026). Covers LLM architectures, prompt engineering, RAG, vector search, and deploying generative AI services.
- **CFA Level I Candidate**, CFA Institute (exam scheduled August 2026). Quantitative methods, equity, fixed income, derivatives, portfolio management.
- **Quantum Computing**, Value-Added Course at VIT Vellore (ongoing).
- **100 Days of Code: The Complete Python Pro Bootcamp** and **The Complete Full-Stack Web Development Bootcamp**, both Udemy (Dr. Angela Yu).

EDUCATION

Vellore Institute of Technology (VIT), Vellore | B.Tech, Data Science & Engineering

2023 – 2027

- **CGPA: 8.84 / 10.0**, consistently in the top percentile of the Data Science cohort.
- **Relevant coursework:** Database Management Systems, Data Structures & Algorithms, Object-Oriented Programming, Probability & Statistics, Machine Learning, Data Mining & Analytics, Operating Systems.
- **Class XII, CBSE: 90.02% (2023) | Class X, CBSE: 93.00% (2021)**

ACHIEVEMENTS & LEADERSHIP

- **Two-Time Hackathon Winner:** First place at the 36-Hour Cyber Forensics Hackathon (Project ARGUS) and at AI Hackathon 2026: The Intelligent Innovation Challenge (Project C.I.T.A.D.E.L.), building both platforms end-to-end under deadline.
- **Research Recognition:** Three patents filed across sustainable computing, agricultural automation, and quantum-inspired modeling, while maintaining an 8.84/10 CGPA.
- **Board Member, IoT in Everyday Life Club, VIT:** Drive strategy, events, and onboarding for a 100+ member technical community; co-organized the "Smart Campus Challenge" hackathon (80+ participants) and led the "IoT Meets AI" workshop series on deploying ML models to edge devices.
- **Mentorship & Community:** Mentored 30+ incoming students through VIT's Freshman Orientation; taught programming and digital literacy to underprivileged high-school students in Vellore district.